

UMAP

Uniform Manifold Approximation and Projection

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What is UMAP?

Goal: project high-dimensional data $X \subset \mathbb{R}^D$ into a low-dimensional space $\mathbb{R}^d$ ($d \ll D$) while *preserving topological structure*.



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Two-phase algorithm

1. **High-D graph:** build a weighted k -NN graph that approximates the manifold topology
2. **Layout:** optimise a low-D embedding whose fuzzy graph matches the high-D one



McInnes, Healy & Melville (2018).

arXiv:1802.03426

Manifold Assumption

Data $\{x_i\} \subset \mathbb{R}^D$ lie on (or near) a compact **Riemannian manifold** $\mathcal{M}$ of *intrinsic* dimension $d \ll D$.

$\Rightarrow$ Only **local** distances are reliable; global Euclidean distances are misleading.



Local balls; geodesic $\neq$ Euclidean



Topological Foundations (I) — Why Topology?

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Simplices — building blocks

0-simplex	point	•
1-simplex	edge	•—•
2-simplex	triangle	△
3-simplex	tetrahedron	

UMAP Algorithm — Two phases

Phase 1 — High-dimensional graph

1. For each x_i , find k nearest neighbours.
2. Define parameter σ_i :

$$\sum_{j=1}^k \exp\left(\frac{-\max(0, d_{ij} - \rho_i)}{\sigma_i}\right) = \log_2 k$$

3. Estimate edge weight $w(i \rightarrow j)$:

$$w_{ij} = \exp\left(\frac{-\max(0, d_{ij} - \rho_i)}{\sigma_i}\right)$$

4. Symmetrise: $_{ij} = w_{ij} + w_{ji} - w_{ij}w_{ji}$

Phase 2 — Low-dimensional layout

5. **Initialise** $Y^{(0)}$ from spectral embedding (eigenvectors of graph Laplacian $\mathcal{L}$).

6. Low-D membership function:

$$\nu_{ij} = \left(1 + a \|y_i - y_j\|^{2b}\right)^{-1}$$

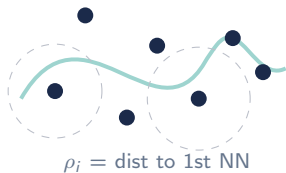
7. Minimise **cross-entropy** via SGD:

$$\mathcal{C}(Y) = \sum_{(i,j)} \underbrace{\bar{w}_{ij} \log \frac{\bar{w}_{ij}}{\nu_{ij}}}_{\text{attractive}} + \underbrace{(1 - \bar{w}_{ij}) \log \frac{1 - \bar{w}_{ij}}{1 - \nu_{ij}}}_{\text{repulsive}}$$

Gradient calculation.

Graph Construction — From Data Points to Fuzzy Graph

(1) Data points



(2) Hard k -NN ($k=2$)



All edges equally weighted — density bias!

(3) Fuzzy graph



Edge width $\propto \bar{w}_{ij}$; density-adaptive

$$\bar{w}_{ij} = w_{ij} + w_{ji} - w_{ij}w_{ji} \text{ (fuzzy union)} \quad w_{ij} = \exp\left(\frac{-\max(0, d_{ij} - \rho_i)}{\sigma_i}\right) \quad \sum_j w_{ij} = \log_2 k \text{ determines } \sigma_i$$

What UMAP (and t-SNE) Plots Do NOT Tell You?

Common misinterpretations

1. **Cluster sizes mean nothing** — large cluster in UMAP plot does not imply more data points or higher density
2. Inter-cluster distances in 2D do not faithfully reflect high-D distances
3. **Random noise doesn't look random** — UMAP can create apparent structure in uniform random data
4. **Hyperparameters dominate** — different $n/\text{min_dist}$ values produce very different plots from same data

Limiting behaviour

For **large** $n_neighbors$ (or perplexity in t-SNE), both methods converge toward **Multi-Dimensional Scaling**: $k \rightarrow n \Rightarrow \text{result} \approx \text{PCA/MDS}$

Non-Euclidean metrics: both UMAP and t-SNE support arbitrary distance functions in $\mathbb{R}^P$ (cosine, Hamming) — not just Euc.distance. **High intrinsic dimension:** if the true manifold dimension d is large ($d > 5$), all projection methods become less informative

tSNE and UMAP comparison

	t-SNE	UMAP
Speed	slow	fast
Global structure	poor	better
Out-of-sample	no	yes
Theory	probabilistic	topology